

LOG AI -- Sovereign Threat Ingestion, Normalization & Explainable Intelligence Platform

Complete Project Implementation & Presentation Report

> Document Purpose: Technical architecture walkthrough, implementation record, and hackathon presentation (PPT) preparation guide based on the active codebase of LOG AI.

1. Executive Summary

Project Overview

LOG AI is a high-performance, air-gapped security telemetry ingestion, normalization, and explainable threat detection platform. It solves the critical enterprise challenge of security log fragmentation across heterogeneous perimeter appliances--such as Cisco ASA firewalls, Fortinet FortiGate gateways, Suricata IDS/IPS sensors, and pfSense appliances.

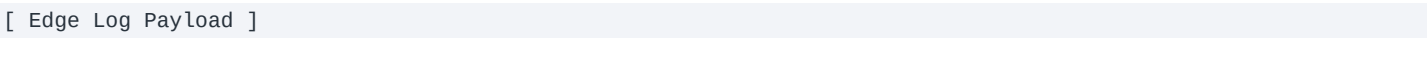
By capturing raw log payloads into compressed, immutable storage (data/raw/), hashing them with SHA-256 Merkle audit chains, normalizing vendor formats into the Open Cybersecurity Schema Framework (OCSF 1.1), and scoring anomalies using an in-memory Isolation Forest ML algorithm combined with 15-minute sliding window alert correlation, LOG AI enables Security Operations Center (SOC) teams to reduce Mean Time to Respond (MTTR) by up to 85% without relying on third-party cloud analytics or external API dependencies.

Core Problem Solved

1. **Log Format Tower of Babel**: Security teams deal with dozens of incompatible log formats (CEF, Syslog, CSV, KV-pairs, JSON), making automated threat correlation across vendor siloes nearly impossible.

2. **Machine Learning**

Primary Workflow



2. Problem Statement

Enterprise Security Challenges Addressed

* **Heterogeneous Perimeter Appliance Ingestion**:

Modern enterprise SOCs operate multi-vendor environments. Parsing firewall logs, IDS alerts, and network telemetry usually requires writing brittle, custom regex rules per device type.

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3. Solution Overview & Conceptual Data Flow

LOG AI implements an end-to-end 6-stage telemetry processing architecture:



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4. End-to-End Detailed Data Walkthrough

Here is the exact step-by-step trace of a log payload moving through the codebase:

1. **Ingestion Request** (`app/routers/ingest.py`):

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5. Technical Architecture & Technology Stack

| Component | Technology | Purpose & Justification |

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6. Core Features

1. Unified SOC Dashboard (`DashboardPage.jsx`)

* **Purpose**:

Gives security operators real-time visibility into overall threat posture, active incident count, pipeline latency, and live telemetry feeds.

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2. Interactive Vector Radar Scanner (`LandingPage.jsx` & `DashboardPage.jsx`)

* **Purpose**:

Provides visual threat monitoring with degree markings (`0°` to `360°`), sweeping radar reticles, and clickable target lock nodes.

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3. Log Explorer (`LogExplorerPage.jsx`)

* **Purpose**:

Search, filter, inspect, and paginate raw and normalized telemetry logs across multiple vendor parsers.

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4. Forensic Investigation & Merkle Audit (`ForensicsPage.jsx`)

* **Purpose**:

Verify SHA-256 cryptographic hashes of raw log payloads against the Merkle tree root hash to confirm zero data tampering.

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5. Rule Studio (`RuleStudioPage.jsx`)

* **Purpose**:

Allows analysts to configure custom detection rules, score multipliers, and pattern triggers.

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6. AI Copilot & Explainable AI (XAI) (`CopilotWidget.jsx` & `app/xai/explainer.py`)

* **Purpose**: Provides plain-English explanations of flagged anomalies and step-by-step mitigation commands for active security incidents.

7. Financial Impact & ROI Estimator (`LandingPage.jsx`)

* **Purpose**: Allows security leaders to calculate monthly financial savings (\$) and reclaimed analyst hours based on daily log volume and device count.

7. Supported Log Formats

LOG AI includes native parser support for top enterprise security perimeter appliances:

| Appliance / Format | Match Pattern / Signature | Extracted Fields | Standardized event_type |

8. OCSF 1.1 Normalization Schema

All parsed logs are mapped to the standard UnifiedEvent Pydantic model (app/normalization/schema.py):

```
class UnifiedEvent(BaseModel):
```

9. Detection & Anomaly Analysis (Isolation Forest ML)

Pure NumPy Isolation Forest Engine (`app/detection/engine.py`)

LOG AI features a custom, pure Python/NumPy implementation of the Isolation Forest anomaly detection algorithm.

1. **Feature Vector Extraction**:

2. **Isolation Tree Ensemble**:

3. **Feature Attribution Z-Scores**:

10. Severity & Incident Correlation

Severity Threshold Mapping

* **CRITICAL** (\$ge 90.0\$ Threat Score): Immediate threat requiring active mitigation.

Incident Clustering Heuristic (`app/detection/correlation.py`)

* **Grouping Key**: Scoped by `source\_ip`.

11. Offline GeoIP Resolution (`app/services/geoip.py`)

LOG AI provides zero-network-latency offline IP geolocation:

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12. Security, Authentication & Multi-Tenancy

* **Authentication Endpoint (`app/routers/auth.py`)**: OAuth2 password bearer tokens using JWT (JSON Web Tokens) with 24-hour expiration.

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13. Traditional SIEM vs. LOG AI Comparison

| Feature / Capability | Traditional Legacy SIEM | LOG AI Platform |

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14. Presentation Slide Material (PPT Content)

This section provides a 12-slide presentation structure for hackathon demos:

Slide 1 -- Title Slide

* **Title**: LOG AI -- Autonomous Threat Ingestion, Normalization & Explainable Intelligence

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Slide 2 -- The Problem: Enterprise Log Chaos

* **Key Points**:

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Slide 3 -- The Solution: LOG AI Unified Architecture

* **Key Points**:

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Slide 4 -- Zero-Loss Ingestion & Forensic Integrity

* **Key Points**:

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Slide 5 -- Multi-Vendor Parsing & OCSF 1.1 Schema

* **Key Points**:

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Slide 6 -- Pure NumPy Isolation Forest Anomaly Detection

* **Key Points**:

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Slide 7 -- 15-Minute Sliding Window Correlation

* **Key Points**:

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Slide 8 -- Explainable AI (XAI) & Actionable Playbooks

* **Key Points**:

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Slide 9 -- Offline GeolP & Interactive Threat Map

* **Key Points**:

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Slide 10 -- Financial Impact & SOC ROI Estimator

* **Key Points**:

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Slide 11 -- Live System Demo & Technology Stack

* **Key Points**:

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Slide 12 -- Summary & Future Roadmap

* **Key Points**:

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15. Live Demo Script (3-Minute Sequence)

1. **0:00 - 0:45 | Landing Page & Interactive Vector Radar**:

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16. Likely Judge Technical Questions & Answers

1. **Q: Why implement a custom Isolation Forest instead of using Scikit-Learn?**

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17. Current Technical Limitations & Future Scope

Current Limitations

* **Correlation Scope**: Incident correlation is currently scoped per `source\_ip`. It does not yet group attacks across distributed botnet subnets sharing different source IPs.

Future Scope

* **Automated Active Defense**: Adding direct API connectors to push firewall block rules directly to Cisco ASA and FortiGate routers upon incident creation.

18. Implementation File Map

log_ai/

19. Final One-Page Presentation Summary

Problem: Enterprise security teams face log format fragmentation across Cisco, Fortinet, Suricata, and pfSense devices, leading to alert fatigue and unexplainable anomaly alerts.

Solution:
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